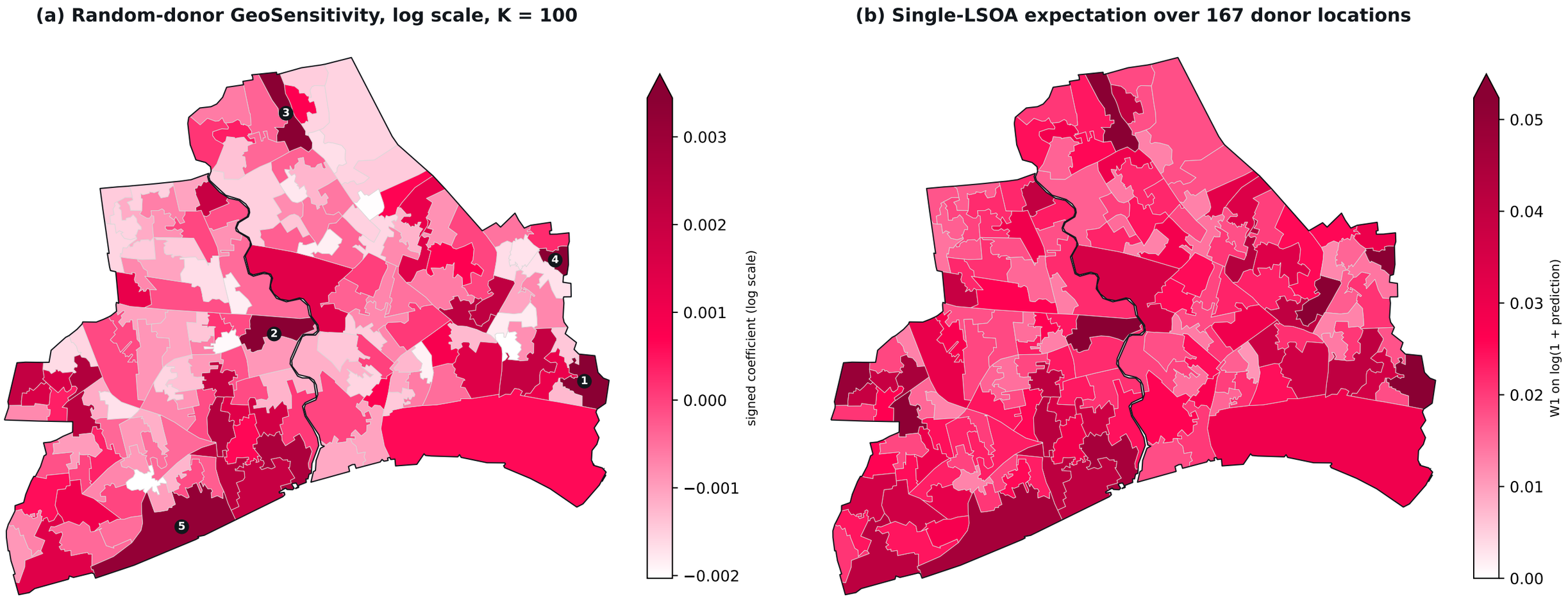


Aligned design (XGBoost on 30 predictors + BNG easting/northing), N=4,000, seed 42. A masked LSOA's coordinates are replaced by those of random other Hull LSOAs (K=100 draws per perturbation); response = W1 on log(1 + prediction), i.e. proportional change. Top 5 numbered. Own sequential scale per panel. Not causal effects.